

Information-Dependent Boundary Response

Conditional Tail Portfolio Decisions: Research Derivations

Local research draft

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Date: 2026-10-08. Status: independently derived finite-dimensional results under the explicit assumptions below; not an externally refereed proof or an exhaustive priority claim. Numerical tests are implementation evidence, not proofs. General continuous-model posterior optimality remains open.

1 Model, conventions, and local expansion

Let g be a fixed proper convex polyhedral function whose effective domain W is a nonempty compact polytope. It includes feasibility and may include a common piecewise-linear turnover penalty. Work on $T = \text{span}(W-W)$, with Euclidean norm. There are finitely many fixed models $j=1, \dots, M$. Define

$$f_{j,t}(w) = \frac{1}{2}w^\top(Q_j + tE_j)w - (\mu_j + tm_j)^\top w + g(w).$$

All Q_j are α -positive on T , $\alpha > 0$, and E_j are bounded. At $t=0$ all models have the same optimizer w_0 . Set $q_j = \mu_j - Q_j w_0$ in the subdifferential of g , $e_j = m_j - E_j w_0$, and

$$a_j(v) = g'(w_0; v) - q_j^\top v, \quad C_j = \{v : a_j(v) = 0\},$$

$$\varphi_j = \sup_{v \in C_j} (e_j^\top v - \frac{1}{2}v^\top Q_j v), \quad h_j(v) = \frac{1}{2}v^\top Q_j v - e_j^\top v + \varphi_j.$$

The tangent cone D of W contains every direction considered. On D , a_j is nonnegative, continuous and piecewise linear. Each C_j is a closed polyhedral cone containing zero. On bounded subsets of D , uniformly as t decreases to zero,

$$R_{j,t}(w_0 + tv) = ta_j(v) + t^2 h_j(v) + O(t^3). \tag{1}$$

Proof of the expansion and its scope

Strong monotonicity gives $\|w_j(t) - w_0\| \leq t\|e_j\|/(\alpha - t\|E_j\|)$. Also, completing the strongly convex quadratic lower bound at w_0 gives $R_{j,t}(w_0) \leq t^2\|e_j\|^2/[2(\alpha - t\|E_j\|)]$. Therefore individual oracle displacements are $O(t)$.

Fix a bounded direction ball. The finitely many inactive affine pieces of g and inactive constraints at w_0 have positive gaps. For all sufficiently small t , none enters on that ball. Hence $g(w_0 + tv) - g(w_0) = t g'(w_0; v)$ exactly, and feasibility is equivalent there to v in D . Expansion of the smooth part is an exact quadratic polynomial plus $t^3 v^\top E_j v / 2$.

A polyhedral error bound gives $\text{dist}(v, C_j) \leq K_j a_j(v)$ on D . On the localized ball the remaining quadratic has a fixed Lipschitz constant. Projecting an individual direction onto C_j changes that quadratic by at most a constant times $a_j(v)$, while its positive penalty is $a_j(v)/t$. For sufficiently small t this penalty is exact. The

oracle improvement thus equals $\hat{t}^2 \phi_j + O(\hat{t}^3)$, proving (1). The projection claim is only for this individual exact-penalty argument; it does not assert that projecting a common action decreases every model's regret.

Constants depend on the fixed finite family, its polyhedral error bounds and α . No uniformity is claimed as M grows, α vanishes, or the constraint description changes.

2 Tail functionals and the implemented optimization

For probability p and tail mass ρ in $(0,1]$, define

$$T_\rho(x; p) = \min_z \{z + \rho^{-1} \sum_j p_j (x_j - z)_+\} = \max_{q \geq 0, \sum_{q_j=1, q_j \leq p_j/\rho} q_j} \sum_j q_j x_j. \quad (2)$$

Introduce nonnegative hinge slacks in the minimization. Its LP dual has one nonnegative multiplier per scenario, bounded above by p_j/ρ ; stationarity in z forces their sum to one. Feasibility and boundedness give equality. This proves convexity, monotonicity, translation equivariance, and positive homogeneity. It also gives $E_p x \leq T_\rho \leq \max$ on the support. Sorting x and filling caps p_j/ρ gives an optimal dual weight, with fractional mass allowed at the threshold. Equal p and $\rho=k/M$ recover the average largest k values.

Splitting an atom into identical copies and splitting its mass leaves (2) unchanged: aggregate dual copy weights for one direction; distribute an original weight proportionally for the reverse direction. Count-based filters need not have this invariance.

For an explicit representation counterexample, take losses $(0,1)$, probabilities $(1/2, 1/2)$, and tail fraction $1/2$. Count Top- k gives 1 and Top- k of probability-times-loss gives $1/2$. Split the zero-loss atom into two identical copies, each of mass $1/4$. The new count $k=\text{ceil}(3/2)=2$ gives $1/2$, and probability-times-loss Top- k gives $1/4$. The genuine probability-tail value remains 1. A change of representation alone therefore changes both count objectives.

For the discontinuous rank-then-reweight rule, use three quadratic losses $((x-\sqrt{3})^2, (x+1)^2, (x-1)^2)$, probabilities $(0.1, 0.8, 0.1)$, and $k=2$. At x approaching zero from below it selects the first and third scenarios and tends to 2; from above it selects the first and second and tends to $11/9$. Deterministic tie-breaking defines a value at zero but does not restore continuity. This example concerns an objective rule, not an assertion that every portfolio instance displays a jump.

Filtering to 95% posterior mass creates a variable number K of retained atoms. An equal-count Top- k rule must then use $k=\text{ceil}(\rho K)$ and effective probability-tail fraction k/K . Keeping the original ρ when K is odd implements a fractional boundary atom instead, a different objective. The implementation has an explicit odd- K regression check.

For joint probabilities $P_{sj}=\alpha_s p_{sj}$, the conditional envelope fixes each row sum α_s and bounds its entries by P_{sj}/ρ . It contains P and is contained in the global cap envelope. Consequently posterior mean $\leq$ conditional tail $\leq$ global tail $\leq$ maximum at each fixed action. This is not a ranking of the true risks of their respective optimizers.

At $\rho=1$ each conditional envelope is the singleton conditional posterior, so the law of total expectation proves $A\text{-Mean}=B\text{-Mean}$. If the full observation density satisfies $p(D|\phi, \eta)=p(D|\phi)$, Bayes' formula cancels the likelihood in the conditional ratio and gives $p(\eta|\phi, D)=p_0(\eta|\phi)$. Conditional posterior and conditional-reference tail objectives then coincide, provided the latter retains the same updated outer marginal. The conclusion is about the entire observation pattern; pairwise scores are not a substitute for its likelihood.

Concavity of likelihood in covariance parameters is not needed for these convex holding problems. After scenarios and their data-dependent masses have been fixed, every regret is convex in w , and a supremum of nonnegative weighted sums is convex. This statement does not make scenario inference globally convex, justify a local likelihood optimizer as a global one, or apply to the discontinuous rank-then-reweight rule.

For any nonempty compact convex envelope U contained in the simplex, convex minimax gives

$$\min_w \max_{q \in U} \sum_j q_j R_j(w) = \max_{q \in U} \{V(\sum_j q_j Q_j, \sum_j q_j \mu_j) - \sum_j q_j V_j\}. \quad (3)$$

The common g is unchanged because weights sum to one. For exact QP solutions, a supergradient is the regret vector of the weighted-QP optimizer. With QP bounds, a feasible action gives $U = \sup_j q_j (f_j(w) - V_j, L)$, and a feasible dual weight gives $L = V_j, L - \sum_j q_j V_j, U$. Evidence penalties add the same linear term to both expressions. Entropic penalties add $-\tau \text{KL}(q||p)$. These are optimization bounds, not frequentist confidence certificates.

The support function of a reverse-KL ball has, for positive b , a maximizing distribution q_j proportional to $b_j / (\nu - x_j)$. Its divergence equation determines $\nu > \max x$. This follows by differentiating the Lagrangian for $\text{KL}(b||q) \leq r$. It is a finite-model adaptation of likelihood robustness, not a calibration theorem for synthetic scenario counts.

3 The probability-scale theorem

Theorem 1 (deterministic changing probabilities). Assume the conditions of Section 1. Fix ρ in $(0, 1]$. Suppose $p(t)$ is a probability vector, $p_j(t)$ converges to p_j^0 , and every $p_j(t)/t$ has an extended nonnegative limit. Define

$$B = \{j : p_j^0 > 0\}, \quad H = \{j : p_j(t)/t \rightarrow \infty\}, \quad S = \{j : p_j(t)/t \rightarrow c_j \in (0, \infty)\}.$$

Then B is nonempty, sum over B of p_j^0 is one, and B is a subset of H . Write $C_H = \text{intersection over } H \text{ of } C_j$. The optimal value satisfies

$$t^{-2} \min_{w \in W} T_\rho(R_t(w); p(t)) \longrightarrow \min_{v \in C_H} \left[T_\rho(\{h_j(v)\}_{j \in B}; p_B^0) + \rho^{-1} \sum_{j \in S} c_j a_j(v) \right]. \quad (4)$$

The rescaled unique optimizing action converges to the unique minimizer on the right. No $O(t^3)$ remainder follows without additional rate assumptions on $p(t)$.

Full proof

Localization. The feasible action w_0 has every regret bounded by $K t^2$, so its tail risk is at most $K t^2$. Since the tail risk dominates the probability mean, choose any b in B . Its probability is bounded below, and every minimizing action w_t has $R_{b,t}(w_t) \leq K t^2 / p_b(t)$. Strong convexity implies $\|w_t - w_b(t)\| = O(t)$. The individual oracle localization above gives $\|w_t - w_0\| = O(t)$. Thus $v_t = (w_t - w_0)/t$ stays in a common compact ball.

Hard directions. Take any convergent subsequence $v_t \rightarrow v$. For j in H , positivity of regret and domination of the mean imply $p_j(t) R_{j,t}(w_t) / t^2 \leq K$. By (1), if $a_j(v) > 0$ then this quantity diverges because $p_j(t)/t$ tends to infinity. Hence v belongs to C_H .

Bounded thresholds, $0 < \rho < 1$. Set $x_{j,t} = R_{j,t}(w_t) / t^2 \geq 0$. For b in B , $x_{b,t}$ is bounded by the mean-risk argument. The total probability outside B goes to zero and is eventually smaller than ρ . The convex threshold objective has a minimizer z_t between zero and max over B of $x_{b,t}$: below zero its slope is negative; above that maximum the mass that can remain above threshold is at most the mass outside B , smaller than ρ , so its slope is positive. Take a subsequence with $z_t \rightarrow z$. These assertions also cover ties by one-sided derivatives.

Lower bound. For b in B , the term $a_b(v_t)/t$ is nonnegative, so $\liminf x_{b,t} \geq h_b(v)$. Continuity and nonnegativity of the hinge give the B contribution at least $\sum p_b^0 (h_b(v) - z)_+ / \rho$. For j in S ,

$$p_j(t)(x_{j,t} - z_t)_+ = \left[\frac{p_j(t)}{t} a_j(v_t) + p_j(t)(h_j(v_t) - z_t + O(t)) \right]_+ \longrightarrow c_j a_j(v).$$

Every remaining hinge contribution is nonnegative and can be discarded. Adding z and minimizing over z yields the lower bound in (4).

For $\rho=1$, T_1 is exactly the mean; use its weighted sum directly. The B terms have the same nonnegative first-order residual, S terms converge to $c_j a_j(v)$, and terms outside B union S have $\liminf$ at least zero because they are weighted regrets. This avoids an incorrect quantile argument at the endpoint.

Recovery. Fix v in C_H . The path w_0+tv is feasible for all small t . For j in H , $a_j(v)=0$. For j in B the normalized losses converge to $h_j(v)$. For H outside B , their losses are bounded and their probabilities vanish. For S , the hinge contribution converges to $c_j a_j(v)$, independently of any bounded z . For the remaining indices $p_j(t)=o(t)$, multiplication of (1) makes their normalized hinge contribution vanish. Evaluating at a minimizer of the limiting B -threshold objective proves the matching upper bound. At $\rho=1$ use the sum. Coercivity ensures the limiting minimizer is finite, so this path supplies a valid recovery action.

Values and optimizers. Every sequence of minimizers is bounded after rescaling. The lower bound and recovery construction identify each accumulation point as a limiting minimizer and give value convergence. Each h_b is α -strongly convex on T . A probability-weighted tail supremum of these quadratics is α -strongly convex, and adding convex a_j terms and restricting to C_H preserves uniqueness. Therefore every subsequence has the same limit. This also establishes uniqueness of the original tail minimizer for small t .

Fixed-probability corollary. If all supported probabilities are fixed positive, $H=B$ and S is empty. The local domain is the intersection of all supported critical cones even for $\rho=1$. Replacing $\max$ by mean or CVaR does not in itself remove this boundary obstruction.

4 Conditional versions: what changes when outer groups disappear

Corollary 2 (positive limiting outer masses). Suppose there are finitely many groups, $\alpha_s(t) > \alpha_s^0 > 0$, and every conditional probability sequence satisfies Theorem 1 with a fixed group tail mass $\rho_s > 0$. Let H_s , S_s and B_s be its conditional sets. Then the scaled conditional objective converges, in minimum and optimizer, to

$$\min_{v \in \cap_s \cap_{j \in H_s} C_{sj}} \sum_s \alpha_s^0 \left[T_{\rho_s}(h_{s,B_s}(v); p_{B_s|s}^0) + \rho_s^{-1} \sum_{j \in S_s} c_{sj} a_{sj}(v) \right]. \quad (5)$$

Proof. The conditional criterion dominates the joint posterior mean, so localization holds using any joint atom with positive limiting mass. Bounded total nonnegative risk implies bounded risk in each positive-mass group. Apply the threshold lower argument group by group to a common convergent v_t . Add finitely many lower bounds. A common recovery direction in all hard cones gives the upper bound simultaneously. At least one positive group contributes uniform strong convexity, proving optimizer convergence.

Proposition 3 (vanishing outer groups with fixed conditional distributions). Allow $\alpha_s(t)$ to vanish, but assume each conditional vector p_s is fixed and positive on its support. Classify groups by $\alpha_s(t)/t$. Groups with that ratio tending to infinity impose every supported C_{sj} . Groups with $\alpha_s(t)/t > c_s$ in $(0, \infty)$ contribute the soft term

$$c_s T_{\rho_s}(\{a_{sj}(v)\}_j; p_s), \quad (6)$$

and groups with $\alpha_s=o(t)$ disappear. The quadratic part comes from groups with $\alpha_s^0 > 0$, using their conditional tail of h . The limiting domain intersects the hard-group cones.

Proof. On bounded directions use homogeneity to write the group's normalized contribution as $(\alpha_s/t) T_{\rho_s}(a_s + t h_s + O(t^2); p_s)$. For a critical group this converges uniformly to (6), since a tail functional is one-Lipschitz in the maximum norm. For a hard group, a nonzero supported a makes its nonnegative tail diverge. On its cone every a vanishes; its surviving second-order contribution is $\alpha_s^0 T_{\rho_s}(h_s; p_s)$. Disappearing groups contribute zero. Localization, lower bounds and recovery then follow as above.

This is a material qualification: a vanishing group's soft term is a **conditional tail of first-order penalties**, not blindly $\sum_j c_{sj} a_{sj}/\rho$. Simultaneously vanishing outer masses and arbitrary changing conditional masses require a more general limiting envelope; that extension remains open here.

5 The exactly interpretable boundary example

For $x \geq 0$, $h, a, s > 0$, let

$$f_{1,t}(x) = \frac{h}{2}x^2 - tax, \quad f_{2,t}(x) = \frac{h}{2}x^2 + sx.$$

At $t=0$ their common optimizer is zero. Their regrets are $R_{1,t} = h(x - ta/h)^2/2$ and $R_{2,t} = h x^2/2 + s x$. If model 2 has probability p_t with $p_t/t > c$, Theorem 1 gives the limiting function

$$\frac{h}{2}(v - a/h)^2 + (cs/\rho)v, \quad v \geq 0.$$

Completing the square yields

$$x_t^*/t \rightarrow [a - cs/\rho]_+/h, \quad T_\rho^*/t^2 \rightarrow a^2/(2h) - [a - cs/\rho]_+^2/(2h). \quad (7)$$

For $p_t/t \rightarrow \infty$ the cone is $\{0\}$; for $p_t = o(t)$ the scaled optimal regret tends to zero. A fixed probability floor prevents that last regime. This example establishes neither the statistical probability rate nor a generic multiasset information lower bound; those are separate questions.

There is also an exact finite- t distinction between a blocked first-order movement and literally zero trading. In this equal-curvature example, the regret curves intersect at

$$x_c = \frac{a^2 t^2}{2h(at + s)}.$$

For fixed $\rho < 1$ and sufficiently small p_t so that $p_t < \rho$ and $1 - p_t \geq \rho$, the optimal action is exactly

$$x_t^* = \max \left\{ x_c, \frac{at - (p_t/\rho)(at + s)}{h} \right\}. \quad (7a)$$

Below x_c the higher regret is model 1's, which alone fills the upper tail, and that regret decreases up to x_c . Above x_c the adverse model receives mass p_t/ρ ; differentiating the resulting weighted quadratic gives the second expression. Convexity proves the formula. Thus the optimizer leaves the $O(t^2)$ kink exactly when

$$p_t < \rho \left[\frac{at}{at + s} - \frac{a^2 t^2}{2(at + s)^2} \right]. \quad (7b)$$

At $\rho=1$ the criterion is a smooth mean instead, and the exact optimum is $[at - p_t(at + s)]_+/h$. Equations (7a)–(7b) explain why a hard limiting cone can coexist with a tiny, nonzero finite-sample holding. A finite numerical kink should not be described as exact zero trading unless the computed rule actually sets it to zero. The log posterior-odds threshold in (7b) is $\log t$ plus a geometry-dependent constant to first order; this motivates the $\log(1/t)$ evidence scale proved in the embedded subclass below.

6 Entropy has a different rare-model threshold

Let $F_\tau = \tau \log \sum_j p_j \exp(R_j/\tau)$. Its variational form is $\sup_q [E_q R - \tau \text{KL}(q||p)]$. Jensen gives $F_\tau \geq E_p R$, providing the same basic localization when a model has nonvanishing probability.

Proposition 4 (fixed temperature). Under Theorem 1, if τ is a fixed positive constant, the scaled entropic minimum has the same limit as (4) with $\rho=1$.

Proof. The mean lower bound and Theorem 1 imply the lower limit. For its recovery direction v in C_H , supported H losses are $O(t^2)$; S probabilities are $O(t)$, with losses $O(t)$; all remaining probabilities are $o(t)$. Thus $E R^2 = O(t^4) + O(t^3) + o(t^3) = o(t^2)$. All losses are $O(t)$, so a uniform Taylor expansion of $\exp(R/\tau)$ and $\log$

gives $F_{\tau} = E R + o(t^2)$ on this recovery path. The value and optimizer claims follow by strong convexity and localization.

Proposition 5 (quadratic temperature, subexponentially rare models). Suppose $\tau = \kappa t^2$ with $\kappa > 0$, all model probabilities are positive, $p(t) > p_0$, and $t \log(1/p_j(t)) > 0$ for every j . Then every limiting minimizing direction belongs to the intersection of all C_j , including those with $p_j = o(t)$. On that intersection the limiting objective is

$$\kappa \log \sum_{j \in B} p_j^0 \exp(h_j(v)/\kappa). \quad (8)$$

Proof. The log-sum-exp dominates any one term: $F_{\tau} \geq R_j + \tau \log p_j$. If $a_j(v) > 0$ along a convergent rescaled direction, division by t gives $\liminf F_{\tau}/t \geq a_j(v) > 0$, contradicting the $O(t^2)$ value at w_0 . On the common cone all first-order penalties vanish. Evaluating a fixed feasible direction gives (8) by continuity of a finite sum. For the lower bound discard nonnegative first-order residuals and retain only B terms. Strong convexity again yields convergence of minimizers.

Polynomially small masses satisfy the premise. Hence CVaR can release an $o(t)$ model while entropy at τ proportional to t^2 still enforces its cone. More generally, the relevant release scale is $\tau \log(1/p_j)$ compared with $t a_j$. If p_j is approximately $\exp(-d_j/t)$ and $\tau = \kappa t^2$, a necessary limiting inequality is $a_j(v) \leq \kappa d_j$. The exact second-order behavior on equality requires prefactor information and is not asserted here.

Warning about an insufficient scaling rule. $\tau/t^2 \rightarrow \infty$ alone does not imply an average-risk limit. For example $\tau = t^{3/2}$ and $p_t = t^2$ still have $\tau \log(1/p_t) = o(t)$, so an adverse first-order boundary can remain hard. Temperature relative to the quadratic scale is not enough; rare probabilities and first-order losses must be considered jointly.

7 Complete-mask evidence and a finite-model sufficient bound

Let model 0 generate independent observations under fixed or parameter-independent masks S_i . Assume all prior masses $b_j > 0$. Put $L_j = \sum_i \log[p_j(Y_i, S_i)/p_0(Y_i, S_i)]$. The posterior probability obeys $p_j(D) \leq (b_j/b_0) \exp(L_j)$. For any λ in $(0, 1)$, let

$$I_{j,\lambda} = - \sum_i \log \int p_0(y|S_i)^{1-\lambda} p_j(y|S_i)^{\lambda} dy.$$

By independence and Markov's inequality,

$$\Pr\{p_j(D) > \epsilon_j\} \leq (b_j/(b_0 \epsilon_j))^{\lambda} \exp(-I_{j,\lambda}). \quad (9)$$

Indeed the event implies $\exp(\lambda L_j) > (\epsilon_j b_0/b_j)^{\lambda}$, and the expectation of that random variable is $\exp(-I)$. A union bound over adverse models supplies a simultaneous statement. Choosing $\epsilon_j = c_j t$ or $t^{1+\gamma}$ links (9) to Theorem 1 in probability when the corresponding probabilities tend to zero. Stronger failure probabilities, e.g. $o(t^2)$ when unscaled regrets are bounded on W , are sufficient to control bad-event contributions to expected regret. Without such control one cannot simply interchange the local limit and expectation.

For zero-mean Gaussians with observed covariance S_0, S_j , the affinity is

$$|S_0|^{-(1-\lambda)/2} |S_j|^{-\lambda/2} |(1-\lambda)S_0^{-1} + \lambda S_j^{-1}|^{-1/2}.$$

This follows by multiplying Gaussian densities and integrating the resulting quadratic exponential. Masks where $S_0 = S_j$ contribute zero information. If only m joint blocks distinguish models, $I = m$ times the joint-block information. The bound explicitly includes prior odds and $\log(1/t)$; it does not replace a random likelihood ratio by its expectation.

Finite nuisance models can be included in the finite union with their prior odds. For a continuous nuisance parameter, the probability and integral bounds require additional local prior-mass and concentration conditions; they are not supplied by merely discretizing a nuisance grid.

8 A matching-order testing subproblem, and its boundary

An information lower bound requires conflicting demands under two observation laws, not merely a small posterior under one truth. Consider the two loss functions in Section 5 with observation laws P_1, P_2 . Assume a rule has $E_1 R_1, t = o(t^2)$ and $E_2 R_2, t = O(t^2)$. Let $A_t = \{x \geq a t / (2h)\}$. Then under model 1, on A_t^c the regret is at least $a^2 t^2 / (8h)$, so $P_1(A_t) \rightarrow 1$. Under model 2, on A_t the regret is at least $s a t / (2h)$, so $P_2(A_t) = O(t)$.

Data processing for KL through this event gives

$$KL(P_1 \| P_2) \geq kl(P_1(A_t), P_2(A_t)) \geq P_1(A_t) \log \frac{1}{P_2(A_t)} - \log 2 \geq (1 - o(1)) \log(1/t) - O(1). \quad (10)$$

If each informative independent block has KL K , mK must therefore be at least this order. For fixed distinct, mutually absolutely continuous Gaussian block laws with positive Chernoff information, a likelihood-ratio test with m proportional to $\log(1/t)$ can make both testing errors $O(t^{-(2+\gamma)})$; choosing each model's oracle action then attains the stated regret requirements. Under model 1 a wrong action costs $O(t^2)$; under model 2 it costs $O(t)$, so these error rates suffice. This proves **matching logarithmic order for this specified statistical decision subproblem**, not a matching constant, general portfolio minimax rate, or optimality of conditional CVaR.

In a portfolio experiment, observation laws and loss matrices may themselves depend on t ; K and Chernoff information can then vanish. Their t -dependence must be retained. The following explicit subclass supplies an embedding with uniformly positive informative-block separation. It is stronger than the abstract testing illustration above but still does not give a general sharp constant.

Theorem 6: a multiasset Gaussian subclass with matching logarithmic information order

Let $d=2k$, $k \geq 5$, and take a long-only budget simplex with cap $u \geq 1/k$. Fix δ in $(0, k)$, $a > 0$ and s in $(0, 1)$. Write $J=11'$ for the k -by- k all-ones matrix and define

$$A = (1 - \delta/k)J + \delta I, \quad B = (4 - \delta/k)J + \delta I, \\ \Sigma_{1,t} = \begin{pmatrix} A & (1 - at)J \\ (1 - at)J & B \end{pmatrix}, \quad \Sigma_{2,t} = \begin{pmatrix} A & (1 + s)J \\ (1 + s)J & B \end{pmatrix}. \quad (11)$$

The objective is half variance, with zero mean. Consider independent, exogenous masks that observe either whole group alone or both groups jointly, with exactly $m=m(t)$ joint observations. Priors b_1, b_2 are fixed and strictly positive. Let ρ in $(0, 1]$ be fixed. The true model is one of these two stated candidates; t indexes a family of decision problems, not an unknown parameter supplied by an evaluation oracle.

(i) For all sufficiently small positive t , both covariance matrices are uniformly positive definite. Their within-group observation laws are identical. At $t=0$ both have the same optimal portfolio $w_0 = (1/k, \dots, 1/k, 0, \dots, 0)$.

(ii) Any decision rules satisfying

$$\mathbb{E}_1 R_{1,t}(\hat{w}_t) = o(t^2), \quad \mathbb{E}_2 R_{2,t}(\hat{w}_t) = O(t^2)$$

must obey

$$m(t)K(t) \geq (1 - o(1)) \log(1/t) - O(1), \quad K(t) = KL(N(0, \Sigma_{1,t}) \| N(0, \Sigma_{2,t})), \quad (12)$$

where $K(t)$ converges to a strictly positive finite constant.

(iii) A posterior model-regret CVaR decision, computed from the complete-mask likelihood, satisfies both requirements when $m(t) \geq C \log(1/t)$ for a sufficiently large fixed C . Thus the required order is logarithmic in this subclass. The constant C is a sufficient Chernoff/affinity constant and is not claimed to match $1/K(0)$.

Proof of (i) and the loss reduction. Orthogonal directions within either group have eigenvalue δ . On the span of the two normalized group indicators, the covariance is k times the two-by-two matrix with diagonals $(1,4)$ and cross entry c . Its determinant is $k^2(4-c^2) > 0$, uniformly for $c=1-at$ or $c=1+s$ and small t . This also proves positive definiteness of A and B . Any within-group mask sees the same marginal in the two models.

Let x be total weight in group two. At a fixed x , uniform weights inside each group simultaneously minimize both objectives. Every departure adds the same nonnegative term δ times the squared distance from the uniform-within-group vector, divided by two. Uniformization respects the cap $u \geq 1/k$. Therefore posterior mean, CVaR and regret minimization all reduce exactly to x in $[0,1]$. After discarding a common constant $1/2$, the two risks are

$$f_{1,t}(x) = \frac{1}{2}(3 + 2at)x^2 - atx, \quad f_{2,t}(x) = \frac{1}{2}(3 - 2s)x^2 + sx.$$

Their oracles are $x_{1,t} = at/(3+2at)$ and $x_{2,t} = 0$, and their exact regrets are

$$R_{1,t}(x) = \frac{1}{2}(3 + 2at)(x - x_{1,t})^2, \quad R_{2,t}(x) = \frac{1}{2}(3 - 2s)x^2 + sx. \quad (13)$$

General nonuniform portfolios have at least these regrets at their group-two mass x . This establishes the boundary behavior and reduction without assuming the decision rule is itself symmetric.

Proof of (ii). Define $E_t = \{x \geq at/6\}$. For small t , $x_{1,t} = at/6 \geq at/12$. Hence on E_t complement, $R_{1,t}$ is at least $c_1 t^2$ for a fixed $c_1 > 0$. The first requirement gives $P_1(E_t) > 1$. On E_t , $R_{2,t} \geq sat/6$, so the second requirement gives $P_2(E_t) = O(t)$. Applying binary KL data processing yields (12), exactly as in (10). All nonjoint masks contribute zero KL, while each independent joint record contributes $K(t)$. Continuity and the strictly distinct, uniformly nonsingular limits of the two Gaussian covariances imply $K(t) > K(0)$ in $(0, \infty)$.

Proof of (iii). Let $p_i(D)$ be the complete-mask posterior. Uniform positive definiteness on a compact small- t interval and distinct limiting Gaussian laws imply their order- $1/2$ affinity is at most $\exp(-I_0)$ for some $I_0 > 0$. Equation (9), including its fixed prior-odds factor, gives

$$P_1\{p_2 > t^{1+\gamma}\} \leq C_0 t^{-(1+\gamma)/2} e^{-mI_0}, \quad P_2\{p_1 > \rho/2\} \leq C_1 e^{-mI_0}.$$

Choose $\gamma > 0$ and $\gamma' > 0$, and C so large that both bounds are $O(t^{\wedge\{2+\gamma\}})$. On the first good event, evaluate the CVaR objective at model 1's oracle $x_{1,t}$. Model 1's regret is zero, model 2's regret is $O(t)$, and $p_2 \leq t^{\wedge\{1+\gamma\}} < \rho$. The resulting tail value is $(p_2/\rho)O(t) = O(t^{\wedge\{2+\gamma\}})$. The optimized tail is no larger and dominates $p_1 R_{1,t}$; eventually $p_1 \geq 1/2$. Consequently the selected action has $R_{1,t} = O(t^{\wedge\{2+\gamma\}})$ on that event.

On the second good event, $p_1 \leq \rho/2$. At $x=0$ model 1 has positive regret and model 2 has zero regret. The right derivative of the posterior CVaR objective is

$$-(p_1/\rho)at + (1 - p_1/\rho)s \geq (s - at)/2 > 0.$$

Convexity then makes $x=0$ optimal. Uniformization is optimal as already proved, so $R_{2,t}$ is exactly zero on that event. Regrets are uniformly bounded over the compact portfolio set and small- t parameter interval. Bad events of probability $O(t^{\wedge\{2+\gamma\}})$ therefore contribute $o(t^2)$ to both expected regrets. This proves the assertions, including the required expectation control. No replacement of a random log likelihood by its mean has been used.

Interpretation and limits. This theorem couples the same covariance matrices to both returns and decision losses, supplies an arbitrary multiasset dimension $2k$, and respects a cap of 0.2 for $k \geq 5$. It is an adaptive boundary-accuracy requirement under two truths, not a lower bound for every tolerance ϵ . The two candidate covariance matrices and their t -indexed decision problem are given to the rule; only the true model identity is unknown. Unknown covariance families, simultaneous estimation of t , continuous nuisance parameters, or a

common optimizer that itself changes with the data are outside this theorem. Conditional CVaR with only one outer group reduces to the rule in (iii); a general advantage of conditional over global tail is not implied.

Proposition 7: very small entropy temperature

Under the local model assumptions, suppose $\tau(t)/t^2 > 0$, all probabilities are positive, $p(t) \rightarrow p_0$, and

$$\tau(t) \log(1/p_j(t))/t^2 \rightarrow D_j < \infty$$

for every model. Then the scaled entropic minimum and optimizer converge to the minimum and minimizer of

$$\max_j \{h_j(v) - D_j\} \quad \text{over} \quad \bigcap_j C_j. \quad (14)$$

Proof. Nonvanishing probabilities and the mean lower bound localize minimizing directions. The one-term inequality $F_{\tau} \geq R_j - \tau \log(1/p_j)$ makes every $a_j(v) > 0$ impossible at a limiting optimizer because the compensating term is only $O(t^2)$. On the common cone, write the normalized objective as a log-sum-exp with temperature τ/t^2 and arguments $R_j/t^2 - \tau \log(1/p_j)/t^2$. Its distance from the maximum argument lies between zero and $(\tau/t^2) \log M$. This vanishes. Nonnegative first-order residuals give the lower bound along varying directions, and a fixed cone direction gives recovery. Strong convexity gives uniqueness. At least one D_j is zero, because a limiting positive-probability model has $D_j = 0$.

In particular, $\tau = t^3$ and polynomial probabilities yield the common-cone minimax quadratic limit. Exponentially tiny probabilities may instead leave finite D_j penalties or fall outside this proposition. The temperature ratio alone is insufficient.

The fixed-temperature conclusion in Proposition 4 also extends to $\tau(t)/t \rightarrow \infty$: on its mean-limit recovery path, the exponential expansion error is $O(E R^2/\tau) = o(t^2)$, since $E R^2 = O(t^3)$ and $\max R/\tau > 0$. This supplies a sufficient average-like regime without incorrectly inferring it from $\tau/t^2 \rightarrow \infty$.

9 Approximation and unresolved claims

Proposition 8 (joint-mass stability, including empty conditional groups). Fix the same finite scenarios and group labels for two joint probability vectors P and P' . Suppose $0 < R_{sj}(w) \leq L$ for every feasible w , and fix $\rho > 0$. Both the global and conditional tail objectives differ uniformly in w by at most

$$\epsilon = (L/\rho) \text{TV}(P, P'), \quad \text{TV}(P, P') = \frac{1}{2} \sum_{s,j} |P_{sj} - P'_{sj}|. \quad (15)$$

This needs no lower bound on individual outer group masses. If each regret is β -strongly convex on the feasible affine hull and w, w' minimize the two objectives, then their optimization-objective excess is at most 2 epsilon and

$$\|w - w'\| \leq 2\sqrt{\epsilon/\beta}. \quad (16)$$

Proof. For the conditional functional, write its threshold form directly in joint masses:

$$F_P(w) = \min_{z_s \in [0, L]} \sum_{s,j} P_{sj} [z_s + \rho^{-1}(R_{sj}(w) - z_s)_+].$$

Every bracket lies in $[0, L/\rho]$. The difference between its expectation under P and P' is therefore at most $(L/\rho) \text{TV}(P, P')$, uniformly in w and all thresholds. Taking minima gives (15). A zero-mass group contributes zero and requires no division by its mass. A single threshold proves the global case. Evaluating each optimum in the other objective gives $F_P(w') - F_{P'}(w) \leq 2 \epsilon$. Strong convexity over the convex feasible set yields $\beta \|w - w'\|^2/2 \leq 2 \epsilon$, proving (16).

This corrects a potentially loose approach that divides conditional-posterior errors by tiny group masses. It applies to common atoms and fixed group definitions. A discrete quadrature and a continuous distribution are mutually singular, so their total variation is generally one; (15) alone cannot certify a continuous quadrature. Scenario refinement and the separate continuous-posterior sampling diagnostic address approximation empirically. A uniform integration error bound over the displayed joint-threshold function class would imply the same 2 epsilon objective-excess argument, but is not established by a high ESS alone.

QP gaps, integration error, and sampling uncertainty are separate. The public implementation reports first-order and weak-duality diagnostics in float64; it is not formally verified interval arithmetic. Scenario refinement can test stability but does not prove that every posterior mode was found. No theorem here gives a frequentist 95% guarantee to a tail objective.

Still open: unrestricted simultaneous outer/inner probability decay; random continuous posterior local limits with nuisance parameters; sharp statistical constants beyond the matching-order multiasset subclass of Theorem 6; data-adaptive strong/weak grouping; growing-dimensional scenario approximation; and a priority analysis against general variational perturbation results. Empirical usefulness of each implemented method is determined separately by the frozen experiments.